

General Description

The RAN650 product family is a series of 5G O-RAN radio units (O-RUs) developed for disaggregated 5G networks in compliance with the O-RAN fronthaul interface specification. This O-RU is designed to support a 7.2x split fronthaul network implementation and is equipped with two 10 GbE fiber fronthaul transport ports. The radio supports up to 4x4 MIMO configuration via external antennae of choice.

The RAN650s provide up to 5 W output power per antenna port. The radio is encased in an IP65 rated outdoor enclosure with 4x N-type antenna ports, 2x 10 GbE SFP+ ports, a GPS antenna port, and a -48 V DC power supply port.

Additionally, the radio unit supports advanced timing synchronization via GPS or IEEE-1588 v2 (PTP) sources and is designed to operate over a temperature range of -33 to +55°C.

Applications

Private campus radio networks
Medium-range base stations
Outdoor 5G access points
Fixed Wireless Access



RAN650 O-RU showing power, data and timing ports

Features

- Frequency Band: n78 (3.3 – 3.8 GHz)
- Carrier bandwidths: Up to 100 MHz*
- Maximum TX output power at each antenna port: +37 dBm
- Modulation schemes up to 256 QAM DL
- MIMO support: 2x2, 4x2, 4x4* (see Benetel feature roadmap)
- SCS: 30 kHz
- Typical power consumption: 100 W
- Sync: GPS, IEEE-1588 v2 (PTP)
- Interfaces: 10 GbE SFP+ fibre
- Operating temperature: -33 up to +55 °C



*Dependent on DU capability and features

Product Details

Interface & External Ports

The Benetel RAN650 is a 5G radio platform configured for outdoor use and designed to support a 7.2x O-RAN split. Interfaces include

- 2 x 10 GbE SFP+ fronthaul transport (one port enabled in the current HW revision)
- Four N-type antenna ports
- GPS antenna port
- DC power port

Sub-Systems and Features

The RAN650 has supporting features to enable flexible implementations. The radio unit supports the following subsystems:

- IEEE 1588v2 PTP - advanced network timing protocol
- GPS Receiver
- Remote software and feature updates via fibre links

Note: For specific O-RU features, including MIMO support, M-plane, alarms, and counters, please refer to Benetel Feature Roadmap. Features such as 4x4 MIMO and carrier bandwidth depend on DU capabilities.

Radio Performance

The RAN650 is equipped with a high-performance radio that supports up to 100 MHz of instantaneous bandwidth and delivers up to 5 W (37 dBm) output power per antenna port.

Table 1: General System Specifications

Parameter	Value
RF Frequency Range	n78 (3300 - 3800 MHz)
Instantaneous Bandwidth	Up to 100 MHz
Power Supply	-48 V
Power Consumption	100 W (typical)
Antenna Configurations	Up to 4T4R
Receiver Noise Figure	4 dB typical
Transmit Output Power	37 +/- 2.5 dBm (at antenna port)
TX adjustable O/P	20 dB
ACLR at max power	47 dBc (typical)*
EVM	3.5% 256 QAM*
Frequency error	100 ppb*
Reference Sensitivity	-98 dBm (typical)*
Modulation Support	256 QAM DL / 64 QAM UL
Sub carrier spacing (SCS)	30 kHz
Time Alignment Error	+/-1.5 us end to end, <400 ns at RU
*3GPP 5G NR Compliance	TS38.104 & TS38.141-1 Release 15

Table 2: Mechanical & Environmental Specifications

Parameter	Conditions
Weight (without mounting bracket)	7.7 kg
Unit Dimensions (LxWxH)	310 x 310 x 108 mm
Unit Volume	10.4L
IP Rating	IP 65
RF Connector	N-type (female)
Grounding	IEC 62368-1
Storage	ETS 300 019-1-1 class 1.2
Transportation	ETS 300 019-1-2 class 2.3
Operational	ETS 300 019-1-4 class 4.1
Certification	CE, EU RED directive 2014/53/EU

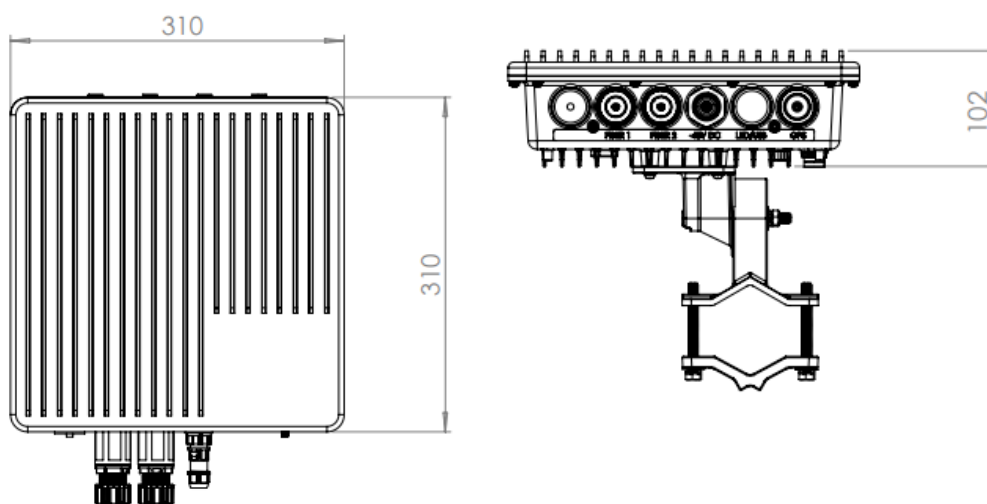
Table 3: Environmental Specifications

Parameter	Value
Environmental Classification	4K2 with extension to +55 °C
Maximum operational ambient temperature (3m/s air-flow, with solar gain, at maximum RF O/P)	+55 °C (+131 °F)
Maximum operational ambient temperature (0m/s air-flow, no solar gain, at maximum RF O/P)	+45 °C (+113 °F)
Minimum operational ambient temperature (including cold start)	-33 °C (-27 °F)

Table 4: Interface & Connectors

Parameter	Signals	Connector Type
10Gbe Ethernet	Optical	SFP+
Transmit/Receive Antenna	RF	N-type
Power Supply Connector	DC -48V	3-Pin Circular female
GPS	RF	SMA

Physical Specification



Note: All dimensions are in millimetres.

Note: The product requires the use of cabled antennas.

Ordering Information

Part Number	Product Description
BNTL-RAN650-1-78AQ CAT-A	5G O-RAN Outdoor Radio Unit 4x4 Band n78 CAT-A, CE certified
BNTL-RAN650-2-78AQ CAT-B	5G O-RAN Outdoor Radio Unit 4x4 Band n78 CAT-B, CE certified
BNTL-RAN650-3-78AQ CAT-A	5G O-RAN Outdoor Radio Unit 4x4 Band n78 CAT-A, CE certified, Mavenir UL1 Compatible

Accessories	Accessory Description
BNTL-AP0056	Fronthaul Fibre Module SFP+ 1310nm 10km
BNTL-AP0059	Weatherproofed aluminium cable gland extension to 10Gb fibre port (optional accessory).

Disclaimer:

The listed part numbers are to be treated as independent product SKUs – select correct option when ordering. It is neither possible to convert between the product SKUs in the field, nor in the factory.

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